



REPUBLIC OF CAMEROON

*Peace – Work – Fatherland*

NORTH WEST REGION

MEZAM DIVISION

BAMENDA I SUB DIVISION



REPUBLIQUE DU CAMEROUN

*Paix – Travail – Patrie*

REGION DE NORD OUEST

DEPARTMENT DU MEZAM

ARRONDESSEMENT DE BAMENDA I

**APPLYING FOR BUILDING PERMIT**

**FOR THE CONSTRUCTION OF A G + 2 INSTITUTIONAL BUILDING**

**BY: MR.ABAMUKONG TABEFOR GESTOPH LOCATED IN BAMENDA AT TAAFI-  
UPSTATION, IN THE MEZAM DIVISION**

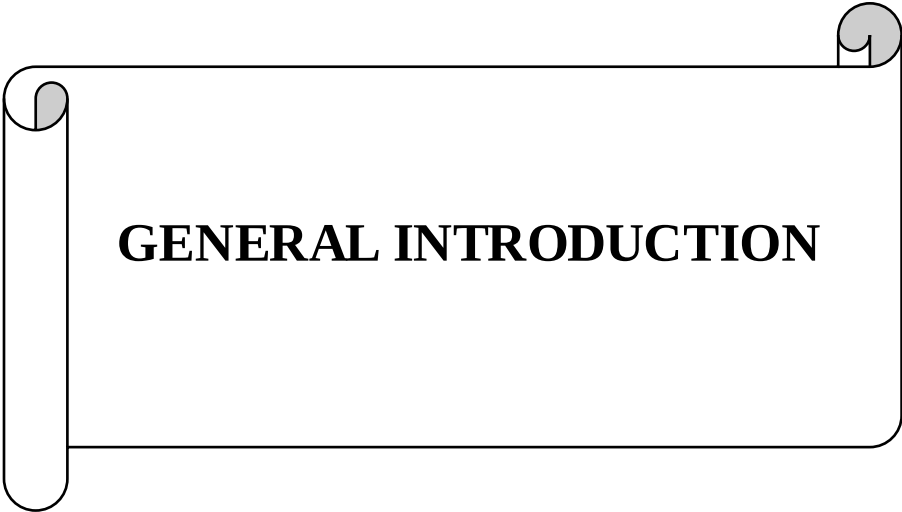
**TECHNICAL SPECIFICATION, BILL OF QUANTITIES AND COST ESTIMATES**

**CONCEPTIONIST: Engr: ACHA BLAISE NYAMBOT**

**APPROVALS**

|                  |              |
|------------------|--------------|
| ARCHITECH'S VISA | COMPANY VISA |
| COUNCIL          | MINH DU      |
| HEALTH           | CITY MAYOR   |





## **GENERAL INTRODUCTION**

## **CONTENTS**

I- DESCRIPTIVE ESTIMATE

II- QUANTITATIVE ESTIMATE

III- CONSTRUCTION PLAN

## **TECHNICAL SPECIFICATIONS**

These specifications are aimed at providing the parties concerned (Engineer, technicians and the client) with necessary technical details with regards to the work to be done. This will give precise information and details as concerning the quality, type and quantity of materials and type of workmanship to be employed in order to realize the project as on the Architectural plans.

### **THE PLOT:**

The plot has topography with a gentle slope that requires not too much excavation before works starts on site, being situated in a private property in a place called **TAAFI, THE MEZAM DIVISION**; see details on the site plan from cadastral service.

### **THE PLAN:**

The Plan as a whole is Residential building having identical rooms, comprising a parlour, kitchen, toilets and verandas, it made of one apartments . This building occupies a total surface area of 600.471m<sup>2</sup> out of the 250m<sup>2</sup> of the plot **TRENCHING:**The main excavations for the foundations (footings and ground beams) are dug about 1.60cm wide and to a depth not **FOUNDATION:** The footings are squared in sizes of 1.50mx1.50m;

Ground beams are of section 20cmx30cm, resting on a 1.2m high blocks concrete filled wall built with cement blocks of 20x20x40cm, details for all reinforcements are given on the structural and foundation plans.

#### **WALL ABOVE:**

All walls shall be of 15x20x40cm cemented blocks built in courses with 2 to 3cm gauged mortar of mixture 1:3

#### **OPENINGS AND LINTELS:**

Some of the doors and windows position shall be marked and frames positioned in their openings (swing) position frames shall be anchored to walls and in the prescribed manner. Woods are treated and seasoned before usage; all openings are with reinforced concrete lintels cast in firm wooden formwork.

#### **CONCRETING:**

All RC mix shall be 1:2:2 with small, big and medium sizes of component aggregates; mixing shall be done manually or mechanically.

#### **ROOFING:**

Rafters shall be framed from selected and treated according to the given design and anchored in position to take purlins and roofing sheets. Rafters are placed at 2m while purlins at 90cm apart.

**FLOOR SCREEDING:**

This shall be done with concrete of 5cm thickness and by a skilled mason to bring the floors to the required levels and finishing.

**SEPTIC TANK and SOAK AWAY PIT:**

A two-chamber tank with an internal dam proof coat and a soak away pit to ensure the sinking of foul water into the ground.

**ELECTRICITY, PLUMBERING and VRD.**

The foreman or project supervisor shall ensure that quality materials and skilled labor are selected for the above (electricity, plumbing and drainage)

**FINISHING:**

Materials (plastering, tiling and painting) for finishing shall be selected and applied to meet the standards as shown on 3 dimensional plans, sections and elevations

**2.2.3. Foundation**

The function of a foundation is to transfer the structural load into the bearing soil without appreciable settlement. It should reach the underlying soil that is free from organic matters, most often; about 30% of the total cost is spent on the foundation.

**Definition: Foundation** is that part of a structure that is usually embedded below the surface of the ground to transmit the load from the super structure to the underlying soil or rock.

#### **2.2.4 Types of foundation**

There are several types of foundation the choice of the foundation type will depend on the following factors: The type of soil on which the foundation is to be built. The type of structure to be realized. The type equipment to be accommodated in the building. The environmental risk factors presented by the site such as; earthquakes, the presence of underlying water, landslides, and erosion etc. Foundation can be classified under two groups namely; shallow foundation and deep foundation.

#### **2.2.5. Footings**

Footings are elements in direct contact with the ground which transmit the load from the posts to the bearing ground. They are usually square, rectangular or circular. Square footings always have square posts of cross section while rectangular footings always have posts of rectangular cross section.

#### **2.2.6 Beam**

A beam is a structural element that has as main role in a building frame to receive all the perpendicular loads to beam axis. Its main mode of

failure is primarily bending and deflection. Once it receives these loads it then transmits those loads plus its self-weight to the columns. The total effect of all the forces acting on the beam is to produce shear forces and bending moments within the beam, that in turn introduce internal stresses, strains and deflection of the beam. Beams are characterized by their manner of support, profile (shape of cross section), length, and their materials.

### ➤ **Classification of beams**

In engineering, beams are of several types based on supports namely;

**-Simply supported beams:** This is a type of beam supported on both ends by supports which are free to rotate and have no moment resistance. **-Fixed beams:** these are beams supported on both ends and restrained from rotation.

**-Over hanging beams:** Here, it consists of a simple beam extending beyond its support on one end.

**-Continuous beams:** Here, the beam extends over more than two supports. A continuous beam is statically indeterminate multi-span beam on hinged supports. The end spans may be cantilever beam. At least one of the supports of continuous beam must be able to develop a reaction along the beam axis which will be seen in the calculation below.

## **2.3. COLUMN /PILLAR:**

There are several types of columns which are used in different parts of structures. Column is a vertical structural member that carries loads mainly in compression. It might transfer loads from a ceiling, floor slab, roof slab, or from a beam, to a floor or foundations, while a pillar also carries load but mostly based on decoration and usually erected/placed at the front of buildings or structure. Reinforce concrete columns is one of the structural members of reinforced concrete frame structures. It is a vertical member which receives load of the slabs, walls, and beams and therefore transmits it directly to the foundation. Columns also have the role to act as vertical supports to suspended members such as beams, and roofs and to transmit the loads of the members down to the foundation. Columns are primary compression members, although they might also have to reduce bending moment transmitted by beams. Columns might be classified based on their categories which shall be seen below;

### **2.3.1. CLASSIFICATION OF COLUMNS**

Columns are classified respectively into 4 main categories as seen below;

- Based on the type of reinforcement.

- Based on the slenderness ratio.

- Based on shape.

- Based on loading.

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**BILL OF QUANTITIES AND COST ESTIMATES FOR THE CONSTRUCTION OF A G+2  
INSTITUTIONAL BUILDING BELONGING TO MR. ABAMUKONG TABEFOR  
GESTOPH, SITUATED AT TAAFI , BAMENDA NORTHWEST REGION OF CAMEROON**

| s/n                                     | Designation                                                        | Unit           | Quantity | Unit Price | Total Price |
|-----------------------------------------|--------------------------------------------------------------------|----------------|----------|------------|-------------|
|                                         |                                                                    |                |          |            |             |
| 101                                     | Levelling and clearing of the site                                 | ls             | 1        | 200,000    | 200,000     |
| 102                                     | Site installation                                                  | ls             | 1        | 150,000    | 150,000     |
| 103                                     | Settingout of the structure                                        | ls             | 1        | 100,000    | 100,000     |
| Sub-Total 100                           |                                                                    |                |          |            | 450,000     |
| A                                       | GROUND FLOOR                                                       |                |          |            |             |
| Lot 200: FOUNDATION WORKS               |                                                                    |                |          |            |             |
| 201                                     | Excavation of foundation trenches/Footings                         | m <sup>3</sup> | 77       | 3,000      | 231,000     |
| 202                                     | 5cm thick lean concrete dosed at 200kg/m3                          | m <sup>3</sup> | 7.00     | 70,000     | 490,000     |
| 203                                     | RC footings dosed at 350kg/m3                                      | m <sup>3</sup> | 20       | 70,000     | 1,400,000   |
| 204                                     | RC for pillars dosed at 350kg/m3                                   | m <sup>3</sup> | 25       | 70,000     | 1,750,000   |
| 205                                     | RC for beams dosed at 350kg/m3                                     | m <sup>3</sup> | 17.44    | 30,000     | 523,200     |
| 206                                     | Frog filled foundation blocks (20x20x40)cm                         | No             | 1599     | 1,000      | 1,599,000   |
| 207                                     | Backfilling and compaction with laterite                           | m <sup>3</sup> | 288.5    | 1,300      | 375,050     |
| 208                                     | Oversite concrete 8cm thick dosed at 200kg/m3                      | m <sup>2</sup> | 30.7     | 3,500      | 107,450     |
| Sub-Total 200                           |                                                                    |                |          |            | 6,475,700   |
| Lot 300: MASONRY WORKS FOR GROUND FLOOR |                                                                    |                |          |            |             |
| 301                                     | 15x20x40cm cement block work for the walls                         | No             | 7988     | 500        | 3,994,000   |
| 302                                     | RC for columns dosed at 350kg/m3                                   | m <sup>3</sup> | 7.83     | 150,000    | 1,174,500   |
| 303                                     | RC for lintels dosed at 350kg/m3                                   | m <sup>3</sup> | 1.5      | 150,000    | 225,000     |
| 304                                     | RC for chain beam dosed at 350kg/m3                                | m <sup>3</sup> | 17       | 150,000    | 2,550,000   |
| 305                                     | RC for staircase dosed at 350kg/m3                                 | m <sup>3</sup> | 2        | 150,000    | 300,000     |
| 306                                     | Plastering/Rendering on walls with cement mortar dosed at 300kg/m3 | m <sup>2</sup> | 1278.4   | 2,000      | 2,556,720   |
| Sub-Total 300                           |                                                                    |                |          |            | 10,800,220  |
| Lot 400: DECKING                        |                                                                    |                |          |            |             |
| 401                                     | Decking blocks of (20x15x50)cm                                     | No             | 3026     | 500        | 1,513,000   |

|                                      |                                                                                           |                |        |         |            |
|--------------------------------------|-------------------------------------------------------------------------------------------|----------------|--------|---------|------------|
| 402                                  | Reinforced concrete for compressive floor slab of 4cm thick dosed at 350kg/m <sup>3</sup> | m <sup>3</sup> | 17     | 20,000  | 340,000    |
| 403                                  | Reinforced concrete for rib beams dosed at 350kg/m <sup>3</sup>                           | m <sup>3</sup> | 3.26   | 20,000  | 65,200     |
| 404                                  | Suffit Boards 0.30x0.03x4.00m                                                             | No             | 81.54  | 4,000   | 326,160    |
| 406                                  | Nails                                                                                     | ls             | 4      | 45,000  | 180,000    |
| 407                                  | Plastering on decking with cement mortar dosed at 300kg/m3                                | m <sup>2</sup> | 416.27 | 2,000   | 832,540    |
| Sub-Total 400                        |                                                                                           |                |        |         | 3,256,900  |
| B                                    | FIRST AND SECOND FLOOR                                                                    |                |        |         |            |
| Lot 500: MASONRY WORKS UPPER FLOOR   |                                                                                           |                |        |         |            |
| 501                                  | (15x20x40)cm cement block work                                                            | No             | 14500  | 500     | 7,250,000  |
| 502                                  | RC for columns dosed at 350kg/m3                                                          | m <sup>3</sup> | 7.83   | 140,000 | 1,096,200  |
| 503                                  | RC for lintels dosed at 350kg/m3                                                          | m <sup>3</sup> | 3      | 140,000 | 420,000    |
| 504                                  | RC for chain beams dosed at 350kg/m3                                                      | m <sup>3</sup> | 34     | 140,000 | 4,760,000  |
| 506                                  | Plastering/Rendering on walls with cement mortar dosed at 300kg/m3                        | m <sup>2</sup> | 2556   | 2,000   | 5,112,000  |
| Sub-Total 500                        |                                                                                           |                |        |         | 18,638,200 |
|                                      |                                                                                           |                |        |         |            |
| Lot 4000: METAL WORKS                |                                                                                           |                |        |         |            |
| 4001                                 | Guard rails on the staircase/Iron balustrades on the veranda and ram way                  | ls             | 16     | 33,000  | 528,000    |
| 4002                                 | window protectors 150x100cm                                                               | ls             | 26     | 50,000  | 1,300,000  |
| 4003                                 | window protectors 120x100cm                                                               | ls             | 16     | 40,000  | 640,000    |
| 4004                                 | window protectors 100x100cm                                                               | ls             | 4      | 30,000  | 120,000    |
| 4005                                 | window protectors 60x60cm                                                                 | ls             | 24     | 25,000  | 600,000    |
| 4006                                 | External doors of 150x210cm with accessories                                              | No             | 4      | 80,000  | 320,000    |
| 4007                                 | External doors of 90x210cm with accessories                                               | No             | 18     | 80,000  | 1,440,000  |
| Sub-Total 4000                       |                                                                                           |                |        |         | 4,948,000  |
| Lot 5000: WOOD WORKS                 |                                                                                           |                |        |         |            |
| 5001                                 | Complete wooden doors 90x210cm with accessories                                           | No             | 18     | 35,000  | 630,000    |
| 5002                                 | Complete wooden doors 80x210cm with accessories                                           | No             |        | 35,000  | 0          |
| Sub-Total 5000                       |                                                                                           |                |        |         | 630,000    |
| Lot 6000: GLASS WORK/ALLUMINIUM WORK |                                                                                           |                |        |         |            |

|                                  |                                                 |                |        |         |                  |
|----------------------------------|-------------------------------------------------|----------------|--------|---------|------------------|
| 6001                             | complete window of 150x110cm with accessories   | No             | 32     | 210,000 | 6,720,000        |
| 6002                             | complete window of 200cmx70cm with accessories  | No             | 2      | 75,000  | 150,000          |
| 6005                             | complete window of 100cmx110cm with accessories | No             | 2      | 20,000  | 40,000           |
| <b>Sub-Total 6000</b>            |                                                 |                |        |         | <b>6,910,000</b> |
| <b>Lot 7000: ELECTRIFICATION</b> |                                                 |                |        |         |                  |
| 7001                             | S and I Conduit pipes Ø40mm                     | Roll           | 15     | 15,000  | 225,000          |
| 7002                             | S and I Conduit pipes Ø30mm                     | Roll           | 15     | 13,000  | 195,000          |
| 7003                             | S & F H07V-U cable, 1.5 mm <sup>2</sup> (Nexan) | Roll           | 12     | 11,000  | 132,000          |
| 7004                             | S & F H07V-U cable, 2.5 mm <sup>2</sup> (Nexan) | Roll           | 10     | 14,000  | 140,000          |
| 7005                             | S & F Wall sockets 2P + T,16A                   | No             | 34     | 1,500   | 51,000           |
| 7007                             | S & F Plastic round cups                        | No             | 64     | 200     | 12,800           |
| 7009                             | Junction box                                    | No             | 12     | 2,400   | 28,800           |
| 7010                             | Complete Distribution fuse box                  | No             | 2      | 6,500   | 13,000           |
| 7012                             | One way switch                                  | No             | 24     | 1,000   | 24,000           |
| 7013                             | Two way switch                                  | No             | 27     | 1,300   | 35,100           |
| 7016                             | Complete incandescent lamps (smart charge)      | No             | 48     | 2,000   | 96,000           |
| 7018                             | Subscription to ENEO                            | ls             | 1      | 400,000 | 400,000          |
|                                  |                                                 |                |        |         |                  |
| <b>Sub-Total 7000</b>            |                                                 |                |        |         | <b>1,352,700</b> |
| <b>Lot 8000: PAINTING</b>        |                                                 |                |        |         |                  |
| 8001                             | Ceiling                                         | m <sup>2</sup> | 302.54 | 800     | 242,032          |
| 8002                             | External walls/internal walls                   | m <sup>2</sup> | 2081.2 | 800     | 1,664,960        |
| 8002                             | Metal/wood works                                | ls             | 1      | 350,000 | 350,000          |
| 8003                             | Skirting on all the walls (35cm high)           | m <sup>2</sup> | 364.22 | 400     | 145,688          |
| <b>Sub-Total 9000</b>            |                                                 |                |        |         | <b>2,402,680</b> |
|                                  |                                                 |                |        |         |                  |

|                             |                                                                                                                                |                |        |       |           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|--------|-------|-----------|
| <b>ROOF WORKS</b>           |                                                                                                                                |                |        |       |           |
| <b>Lot 9000: ROOF WORKS</b> |                                                                                                                                |                |        |       |           |
| 9001                        | (5x15)cm hard wood for the rafters, braces and struts                                                                          | No             | 300    | 5,000 | 1,500,000 |
| 9002                        | Purlins and battens of 6cmx7cm for the roof, gable ends and fascia boards                                                      | No             | 800    | 1,200 | 960,000   |
| 9003                        | Complete ceiling with plywood 4cm thick on the entire ceiling surface alongside hangers, hardwood joists and modeled linnings. | m <sup>2</sup> | 302.54 | 8,000 | 2,420,320 |

|                       |                                                              |                |      |         |                  |
|-----------------------|--------------------------------------------------------------|----------------|------|---------|------------------|
| 9004                  | Coated tole Bac roofing sheet 6/10, 6m Long with accessories | m <sup>2</sup> | 36.9 | 5,500   | 202,950          |
| 9005                  | Top ridging 4ml                                              | No             | 25   | 4,000   | 100,000          |
| 9006                  | Gutter                                                       | ls             | 18   | 4,000   | 72,000           |
| 9007                  | Nails and other accessories                                  | ls             | 1    | 100,000 | 100,000          |
| <b>Sub-Total 9000</b> |                                                              |                |      |         | <b>5,355,270</b> |

|                                                     |                                                  |                |        |       |                  |
|-----------------------------------------------------|--------------------------------------------------|----------------|--------|-------|------------------|
| <b>Lot 10000: WALL AND FLOOR COVERINGS (TILING)</b> |                                                  |                |        |       |                  |
| 10001                                               | Floor tiles for rooms, toilets lobby and gallery | m <sup>2</sup> | 302.54 | 8,000 | 2,420,320        |
| 10002                                               | Wall tiles (toilet walls) 2m high                | m <sup>2</sup> | 99.12  | 5,500 | 545,160          |
| <b>Sub-Total 10000</b>                              |                                                  |                |        |       | <b>2,965,480</b> |
| <b>Lot 11000: PLUMBING WORKS</b>                    |                                                  |                |        |       |                  |
| <b>Water Supply</b>                                 |                                                  |                |        |       |                  |
| 11001                                               | Flexible green pipes 1/2"                        | No             | 60     | 2,500 | 150,000          |
| 11002                                               | Green Elbows 1/2"                                | No             | 40     | 800   | 32,000           |
| 11003                                               | Green tees 1/2"                                  | No             | 30     | 800   | 24,000           |
| 11004                                               | GI elbows 1/2"                                   | No             | 25     | 150   | 3,750            |
| 11005                                               | GI nipples 1/2"                                  | No             | 23     | 150   | 3,450            |
| 11006                                               | GI suckets 1/2"                                  | No             | 21     | 150   | 3,150            |
| 11007                                               | GI male/female elbows 1/2                        | No             | 23     | 150   | 3,450            |
| 11008                                               | Brass stop corks 1/2"                            | No             | 50     | 1,000 | 50,000           |
| 11009                                               | Herms 1/2" (bundles)                             | No             | 5      | 1,000 | 5,000            |
| 11010                                               | Geba joint (tins)                                | No             | 10     | 3,500 | 35,000           |
| 11011                                               | GI Plugs 1/2"                                    | No             | 56     | 150   | 8,400            |
| <b>Sub-Total</b>                                    |                                                  |                |        |       | <b>318,200</b>   |
| <b>Plumbing System</b>                              |                                                  |                |        |       |                  |
| 11012                                               | Tangit gum (tins)                                | No             | 10     | 7,000 | 70,000           |
| 11013                                               | PVC Pipes 100mm                                  | No             | 30     | 3,000 | 90,000           |
| 11014                                               | PVC Pipes 63mm                                   | No             | 25     | 1,500 | 37,500           |
| 11015                                               | Elbow 100mm                                      | No             | 20     | 700   | 14,000           |
| 11016                                               | "Y" 100mm                                        | No             | 20     | 1,000 | 20,000           |
| 11017                                               | Tees 63mm                                        | No             | 13     | 500   | 6,500            |
| 11018                                               | Elbow 63mm                                       | No             | 15     | 400   | 6,000            |
| 11019                                               | Plugs 63mm                                       | No             | 12     | 500   | 6,000            |
| 11020                                               | PVC Reducers 63mmx32mm                           | No             | 10     | 400   | 4,000            |
| 11021                                               | Shower drain                                     | No             | 10     | 500   | 5,000            |
| <b>Sub-Total</b>                                    |                                                  |                |        |       | <b>259,000</b>   |

| Installation and Services  |                                        |    |    |         |           |
|----------------------------|----------------------------------------|----|----|---------|-----------|
| 11022                      | English toilets (WC)                   | No | 12 | 50,000  | 600,000   |
| 11023                      | Shower strap corks                     | No | 12 | 2,000   | 24,000    |
| 11024                      | Basin taps                             | No | 12 | 2,500   | 30,000    |
| 11025                      | Kitchen sinks                          | No | 3  | 20,000  | 60,000    |
| 11026                      | Sink taps                              | No | 3  | 2,500   | 7,500     |
| 11027                      | Bath mirrors with lights               | No | 6  | 5,000   | 30,000    |
| 11028                      | Towel hanger                           | No | 6  | 1,500   | 9,000     |
| 11029                      | Tissue holder                          | No | 6  | 1,500   | 9,000     |
| 11030                      | Toilet brush                           | No | 6  | 1,000   | 6,000     |
| 11031                      | Toilet taps                            | No | 6  | 1,500   | 9,000     |
| 11033                      | Showers                                | No | 0  | 15,000  | 0         |
| 11034                      | Screws (packets)                       | No | 6  | 1,000   | 6,000     |
| 11035                      | Pegs (sherins) (packets)               | No | 6  | 1,500   | 9,000     |
| Sub-Total                  |                                        |    |    |         | 799,500   |
| Sub-Total 11000            |                                        |    |    |         | 1,376,700 |
|                            |                                        |    |    |         |           |
| Lot 12000: DRAINAGE SYSTEM |                                        |    |    |         |           |
| 12001                      | Rain water pipes with accessories (4m) | No | 16 | 6000    | 96000     |
| 12002                      | Soakaway                               | ls | 1  | 200,000 | 200,000   |
| 12003                      | Septic tank                            | ls | 1  | 600,000 | 600,000   |
| Sub-Total 12000            |                                        |    |    |         | 896,000   |

| SUMMARY |                            |            |
|---------|----------------------------|------------|
| 100     | PRELIMINARY WORKS          | 450,000    |
| A       | GROUND FLOOR               |            |
| 200     | FOUNDATION WORKS           | 6,475,700  |
| 300     | MASONRY WORKS              | 10,800,220 |
| 400     | DECKING                    | 3,256,900  |
| A       | UPPER FLOOR                | 18,638,200 |
| 500     | MASONRY WORKS              | 18,638,200 |
| 4000    | METAL WORKS                | 4,948,000  |
| 5000    | WOOD WORKS                 | 630,000    |
| 6000    | GLASS WORK/ALLUMINIUM WORK | 6,910,000  |
| 7000    | ELECTRIFICATION            | 1,352,700  |
| 8000    | PAINTING                   | 2,402,680  |

|                                                                             |                                   |            |
|-----------------------------------------------------------------------------|-----------------------------------|------------|
| 9000                                                                        | ROOF WORKS                        | 5,355,270  |
| 10000                                                                       | WALL AND FLOOR COVERINGS (TILING) | 2,965,480  |
| 11000                                                                       | PLUMBING WORKS                    | 1,376,700  |
| 12000                                                                       | DRAINAGE SYSTEM                   | 896,000    |
|                                                                             |                                   |            |
| GRAND TOTAL                                                                 |                                   | 60,206,580 |
| SIXTY MILLION, TWO-HUNDRED AND SIX THOUSANDS FIVE HUNDRED AND EIGHTY FRANCS |                                   |            |

| BILL OF QUANTITIES AND COST ESTIMATES FOR THE CONSTRUCTION OF ADMINISTRATIVE BLOCK |                                                                                |                |          |            |                  |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------|----------|------------|------------------|
| S/N                                                                                | DESIGNATION                                                                    | UNIT           | QUANTITY | UNIT PRICE | TOTAL PRICE      |
| <b>Lot 100: PREPARATORY WORKS</b>                                                  |                                                                                |                |          |            |                  |
| 101                                                                                | Studies and project drawings                                                   | ls             | 1        | 200000     | 200,000          |
| 102                                                                                | Site installation                                                              | ls             | 1        | 50000      | 50,000           |
| 103                                                                                | Levelling of the site                                                          | m <sup>2</sup> | 277      | 250        | 69,250           |
| 104                                                                                | Settingout of the structure                                                    | ls             | 1        | 100000     | 100,000          |
| <b>Sub-Total 100</b>                                                               |                                                                                |                |          |            | <b>419,250</b>   |
| <b>Lot 200: FOUNDATION WORKS</b>                                                   |                                                                                |                |          |            |                  |
| 201                                                                                | Excavation of foundation trenches                                              | m <sup>3</sup> | 32.49    | 3000       | 97,470           |
| 202                                                                                | 5cm thick lean concrete dosed at 200kg/m3                                      | m <sup>3</sup> | 2.80     | 65000      | 182,000          |
| 203                                                                                | Random rubble masonry foundation or frog filled foundation blocks (20x20x40)cm | ls             | 680      | 500        | 340,000          |
| 204                                                                                | RC for foundation pillars and beams dosed at 350kg/m3                          | m <sup>3</sup> | 1.3      | 145000     | 188,500          |
| 205                                                                                | Backfilling and compaction with laterite                                       | m <sup>3</sup> | 12.35    | 1300       | 16,055           |
| 206                                                                                | 10cm thick hardcore                                                            | m <sup>3</sup> | 10.63    | 4700       | 49,961           |
| 207                                                                                | Oversite concrete 6cm thick dosed at 200kg/m3                                  | m <sup>2</sup> | 7.63     | 4500       | 34,335           |
| 208                                                                                | 5cm thick sandscreed                                                           | m <sup>2</sup> | 119.49   | 1500       | 179,235          |
|                                                                                    |                                                                                |                |          |            | -                |
| <b>Sub-Total 200</b>                                                               |                                                                                |                |          |            | <b>1,087,556</b> |
| <b>Lot 300: MASONRY WORKS</b>                                                      |                                                                                |                |          |            |                  |
| 301                                                                                | (15x20x40)cm blocks                                                            | No             | 5300     | 300        | 1,590,000        |
| 302                                                                                | RC for pillars, beams and lintels dosed at 350kg/m3                            | m <sup>3</sup> | 2.63     | 145000     | 381,350          |

|                             |                                                                                                                                |                |        |        |                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|--------|--------|------------------|
| 303                         | Plastering/Rendering with cement mortar dosed at 300kg/m3                                                                      | m <sup>2</sup> | 519.23 | 2500   | 1,298,075        |
| <b>Sub-Total 300</b>        |                                                                                                                                |                |        |        | <b>3,269,425</b> |
| <b>Lot 400: ROOF WORKS</b>  |                                                                                                                                |                |        |        |                  |
| 401                         | (5x15)cm hard wood for the rafters, braces and struts                                                                          | No             | 170    | 2500   | 425,000          |
| 402                         | Purlins and battens of 5cmx7cm for the roof, gable ends and fascia boards                                                      | No             | 170    | 1200   | 204,000          |
| 403                         | Complete ceiling with plywood 4cm thick on the entire ceiling surface alongside hangers, hardwood joists and modeled linnings. | m <sup>2</sup> | 150    | 3500   | 525,000          |
| 404                         | Coated tole Bac roofing sheet 6/10, 6m Long with accessories                                                                   | m <sup>2</sup> | 119    | 5800   | 690,200          |
| 405                         | Verandah ceiling with tole lise                                                                                                | m <sup>2</sup> | 20.08  | 3500   | 70,280           |
| 406                         | Fascia boards 2ml                                                                                                              | U              | 25     | 3000   | 75,000           |
| 407                         | Top ridging and valley gutter sheets , 2ml                                                                                     | U              | 25     | 3500   | 87,500           |
| 408                         | Nails and other accessories                                                                                                    | ls             | 1      | 120000 | 120,000          |
| <b>Sub-Total 400</b>        |                                                                                                                                |                |        |        | <b>2,196,980</b> |
| <b>Lot 500: METAL WORKS</b> |                                                                                                                                |                |        |        |                  |
| 501                         | Iron balustrades on verandahs                                                                                                  | ls             | 1      | 30000  | 30,000           |
| 502                         | Complete rails for curtains                                                                                                    | ls             | 1      | 20000  | 20,000           |
| 503                         | External doors of 100x220cm with accessories                                                                                   | No             | 0      | 0      | -                |
| 504                         | Doors 90cmx210cm                                                                                                               | No             | 6      | 55000  | 330,000          |
| 505                         | Window protectors 200x120cm with accessories                                                                                   | No             | 0      | 0      | -                |
| 506                         | Window protectors 150x210cm with accessories                                                                                   | No             | 9      | 30000  | 270,000          |
| 507                         | Window protectors 60x60cm with accessories                                                                                     | No             | 4      | 15000  | 60,000           |
| <b>Sub-Total 500</b>        |                                                                                                                                |                |        |        | <b>710,000</b>   |
| <b>Lot 600: WOOD WORKS</b>  |                                                                                                                                |                |        |        |                  |

|                                            |                                                 |      |    |       |                |
|--------------------------------------------|-------------------------------------------------|------|----|-------|----------------|
| 601                                        | Complete wooden doors 90x220cm                  | No   | 0  | 0     | -              |
| 602                                        | Complete wooden doors 70x220cm                  | No   | 2  | 25000 | 50,000         |
| 603                                        | Complete wardrobes                              | No   | 0  | 0     | -              |
| 604                                        | Complete kitchen shelve                         | ls   | 1  | 30000 | 30,000         |
| <b>Sub-Total 600</b>                       |                                                 |      |    |       | <b>80,000</b>  |
| <b>Lot 700: GLASS WORK/ALLUMINIUM WORK</b> |                                                 |      |    |       |                |
| 701                                        | Windows of 200x120cm with accessories           | No   |    |       | -              |
| 702                                        | Windows of 150x210cm with accessories           | No   | 8  | 45000 | 360,000        |
| 703                                        | Windows of 60x60cm with accessories             | No   | 4  | 21000 | 84,000         |
| 704                                        |                                                 |      |    |       | -              |
| <b>Sub-Total 700</b>                       |                                                 |      |    |       | <b>444,000</b> |
| <b>Lot 800: ELECTRIFICATION</b>            |                                                 |      |    |       |                |
| 801                                        | S and I Conduit pipes Ø11mm <sup>2</sup>        | roll | 3  | 10000 | 30,000         |
| 802                                        | S & F H07V-U cable, 1.5 mm <sup>2</sup> (Nexan) | roll | 6  | 9000  | 54,000         |
| 803                                        | S & F H07V-U cable, 2.5 mm <sup>2</sup> (Nexan) | roll | 4  | 14000 | 56,000         |
| 805                                        | S & F Wall sockets 2P + T,16A                   | No   | 6  | 1200  | 7,200          |
| 807                                        | S & F Plastic round cups                        | No   | 15 | 200   | 3,000          |
| 808                                        | Power sockets 2L+E                              | No   | 6  | 2300  | 13,800         |
| 809                                        | Junction box                                    | No   | 2  | 2400  | 4,800          |
| 810                                        | Complete Distribution fuse box                  | No   | 2  | 6500  | 13,000         |
| 811                                        | Antenna socket                                  | No   | 4  | 1350  | 5,400          |
| 812                                        | One way switch                                  | No   | 6  | 1200  | 7,200          |
| 813                                        |                                                 | No   |    |       |                |

|                                                    |                                               |                |        |        |                  |
|----------------------------------------------------|-----------------------------------------------|----------------|--------|--------|------------------|
| 814                                                | Two way switch                                | No             | 5      | 1300   | 6,500            |
| 815                                                |                                               |                |        |        | -                |
| 816                                                | Complete incandescent lamps (smart charge)    | No             | 10     | 1300   | 13,000           |
|                                                    |                                               |                |        |        | -                |
|                                                    |                                               |                |        |        | -                |
| 819                                                | Subscription to ENEO                          | ls             | 1      | 120000 | 120,000          |
| <b>Sub-Total 800</b>                               |                                               |                |        |        | <b>333,900</b>   |
| <b>Lot 900: PAINTING</b>                           |                                               |                |        |        |                  |
| 901                                                | Ceiling                                       | m <sup>2</sup> | 40     | 1000   | 40,000           |
| 902                                                | External walls/internal walls                 | m <sup>2</sup> | 55     | 1000   | 55,000           |
| 903                                                | Metal and wood works                          | m <sup>2</sup> | 15.35  | 1200   | 18,420           |
| 904                                                | Skirting on all the walls (35cm high)         | m <sup>2</sup> | 32     | 1000   | 32,000           |
| <b>Sub-Total 900</b>                               |                                               |                |        |        | <b>145,420</b>   |
| <b>Lot 1000: WALL AND FLOOR COVERINGS (TILING)</b> |                                               |                |        |        |                  |
| 1001                                               | Floor tiles                                   | m2             | 119.45 | 9500   | 1,134,775        |
| 1002                                               | Wall tiles (toilet and kitchen walls) 2m high | m2             | 35     | 7500   | 262,500          |
| <b>Sub-Total 1000</b>                              |                                               |                |        |        | <b>1,397,275</b> |
| <b>Lot 1100: SURFACE DRAINAGE</b>                  |                                               |                |        |        |                  |
|                                                    |                                               |                |        |        |                  |
| 1102                                               | Roof gutter                                   | ml             | 50     | 2000   | 100,000          |
|                                                    |                                               |                |        |        | -                |
| <b>Sub-Total 1100</b>                              |                                               |                |        |        | <b>100,000</b>   |
| <b>Lot 1200: PLUMBING WORKS</b>                    |                                               |                |        |        |                  |
| <b>Water Supply</b>                                |                                               |                |        |        |                  |
| 1201                                               | Flexible green pipes 1/2"                     | No             | 7      | 2500   |                  |

|                        |                           |    |    |      |               |
|------------------------|---------------------------|----|----|------|---------------|
|                        |                           |    |    |      | 17,500        |
| 1202                   | Green Elbows 1/2"         | No | 15 | 800  | 12,000        |
| 1203                   | Green tees 1/2"           | No | 15 | 800  | 12,000        |
| 1204                   | GI elbows 1/2"            | No | 15 | 150  | 2,250         |
| 1205                   | GI nipples 1/2"           | No | 15 | 150  | 2,250         |
| 1206                   | GI suckets 1/2"           | No | 15 | 150  | 2,250         |
| 1207                   | GI male/female elbows 1/2 | No | 15 | 150  | 2,250         |
| 1208                   | Brass stop corks 1/2"     | No | 4  | 1000 | 4,000         |
| 1209                   | Herms 1/2" (bundles)      | No | 4  | 1000 | 4,000         |
| 1210                   | Geba joint (tins)         | No | 4  | 3500 | 14,000        |
| 1211                   | GI Plugs 1/2"             | No | 18 | 150  | 2,700         |
| <b>Sub-Total</b>       |                           |    |    |      | <b>75,200</b> |
| <b>Drainage System</b> |                           |    |    |      |               |
| 1212                   | Tangit gum (tins)         | No | 5  | 7000 | 35,000        |
| 1213                   | PVC Pipes 100mm           | No | 9  | 3000 | 27,000        |
| 1214                   | PVC Pipes 63mm            | No | 10 | 1500 | 15,000        |
| 1215                   | Elbow 100mm               | No | 13 | 700  | 9,100         |
| 1216                   | Tees 100mm                | No | 9  | 1000 | 9,000         |
| 1218                   | Tees 63mm                 | No | 19 | 500  | 9,500         |
| 1219                   | Elbow 63mm                | No | 12 | 400  | 4,800         |
| 1220                   | Plugs 63mm                | No | 12 | 500  | 6,000         |
| 1221                   | PVC Reducers 63mmx32mm    | No | 10 | 400  | 4,000         |
| 1222                   | Shower drain              | No | 3  | 500  | 1,500         |

|                                  |                                  |    |   |       |                  |
|----------------------------------|----------------------------------|----|---|-------|------------------|
| <b>Sub-Total</b>                 |                                  |    |   |       | <b>120,900</b>   |
| <b>Installation and Services</b> |                                  |    |   |       |                  |
| 1223                             | English toilets (WC)             | No | 3 | 50000 | 150,000          |
| 1224                             | Shower strap corks               | No | 3 | 2000  | 6,000            |
| 1225                             | Basin taps                       | No | 3 | 3000  | 9,000            |
| 1226                             | Kitchen sinks                    | No | 1 | 25000 | 25,000           |
| 1227                             | Sink taps                        | No | 1 | 3000  | 3,000            |
| 1228                             | Bath mirror with lights          | No | 3 | 10000 | 30,000           |
|                                  |                                  |    |   |       | -                |
| 1231                             | Tissue holder                    | No | 4 | 1500  | 6,000            |
| 1232                             | Toilet brush                     | No | 4 | 1500  | 6,000            |
| 1233                             | Toilet taps                      | No | 4 | 1500  | 6,000            |
| 1234                             | Water heater (elcteric) 15litres | No | 1 | 45000 | 45,000           |
|                                  |                                  |    |   |       | -                |
| 1336                             | Screws (packets)                 | No | 4 | 1000  | 4,000            |
| 1337                             |                                  |    |   |       | -                |
| 1338                             |                                  |    |   |       | -                |
| <b>Sub-Total</b>                 |                                  |    |   |       | <b>290,000</b>   |
| <b>Sub-Total 1200</b>            |                                  |    |   |       | <b>486,100</b>   |
| <b>SUMMARY</b>                   |                                  |    |   |       |                  |
| <b>100</b>                       | <b>PREPARATORY WORKS</b>         |    |   |       | <b>419,250</b>   |
| <b>200</b>                       | <b>FOUNDATION WORKS</b>          |    |   |       | <b>1,087,556</b> |
| <b>300</b>                       | <b>MASONRY WORKS</b>             |    |   |       | <b>3,269,425</b> |

|                                                                                                                             |                                          |                   |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------|
| <b>400</b>                                                                                                                  | <b>ROOF WORKS</b>                        | <b>2,196,980</b>  |
| <b>500</b>                                                                                                                  | <b>METAL WORKS</b>                       | <b>710,000</b>    |
| <b>600</b>                                                                                                                  | <b>WOOD WORKS</b>                        | <b>80,000</b>     |
| <b>700</b>                                                                                                                  | <b>GLASS WORK/ALLUMINIUM WORK</b>        | <b>444,000</b>    |
| <b>800</b>                                                                                                                  | <b>ELECTRIFICATION</b>                   | <b>333,900</b>    |
| <b>900</b>                                                                                                                  | <b>PAINTING</b>                          | <b>145,420</b>    |
| <b>1000</b>                                                                                                                 | <b>WALL AND FLOOR COVERINGS (TILING)</b> | <b>1,397,275</b>  |
| <b>1100</b>                                                                                                                 | <b>SURFACE DRAINAGE</b>                  | <b>100,000</b>    |
| <b>1200</b>                                                                                                                 | <b>PLUMBING WORKS</b>                    | <b>486,100</b>    |
| <b>GRAND TOTAL</b>                                                                                                          |                                          | <b>10,669,906</b> |
| <b>THIS ESTIMATE IS FINALLY CLOSED AT TEN MILLION SIX HUNDRED AND SIXTY NINE THOUSANDS NINE HUNDRED AND SIX FRANCS CFA.</b> |                                          |                   |



**ARCHITECTURAL PLAN**



